Audio-Visual Cross-Alignment for Multimodal Deepfake Detection

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Introduction

- **Multimodal Deepfake Detection**

Deepfakes can manipulate audio, facial appearance, facial motion, or their combination.

- **Three Sources of Evidence**

VIDEO	AUDIO	CROSS-MODAL
Facial motion	Voice characteristics	Lip–speech synchronization
Lip dynamics	Speech dynamics	Temporal correspondence
Facial artifacts	Acoustic artifacts	Audio–visual consistency

- **The Challenge**

Audio and video may appear realistic individually while their relationship remains inconsistent.

- **Our Focus**

Detecting deepfakes through audio–visual cross-alignment.

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Literature Review

Paper	Novelty	Result	Limitation
FlowFake (2026)	LTC network models continuous-time audio dynamics across multiple scales	75.29% FakeOrReal→ASVspoof; 90.41% zero-shot WaveFake	Performance drops under domain shift
BioLip (2026)	64 perioral landmarks + displacement, velocity, acceleration & jerk	0.905 AUC English; 0.828±0.013 on 7-language PolyglotFake	Sensitive to compression/noise
AVFF	Self-supervised AV learning + complementary masking	98.6% accuracy; 99.1% AUC on FakeAVCeleb	Sensitive to asynchronous/mismatched AV streams
MRDF	Cross- and within-modality regularization with AV Transformer	94.05% accuracy; 92.43% AUC on FakeAVCeleb	Limited broad cross-dataset evidence

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Literature Review Contd.

Paper	Novelty	Result	Limitation
Contextual Cross-Modal Attention (2024)	Multi-sequence attention for detection + temporal localization	0.968 AUC / 0.956 ACC on LAV-DF	Modality/distribution gap remains
NPVForensics	Non-critical phoneme-viseme correlation + alignment	98.5% cross-manipulation AUC; 92.5% cross-dataset AUC	Audio-dependent; cross-lingual performance untested
AVSCNet (2026)	Separate synchronization + content-consistency branches	98.30% AUC / 98.20% AP on FakeAVCeleb	24.6M parameters; computationally costly
X-AVDT (2026)	Diffusion inversion + generator-internal AV cross-attention	95.29% avg. AUROC on MMDF	~1 min/clip; expensive inference

Research Gaps

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01 | Isolated Modality Evidence - FlowFake and BioLip provide lightweight audio/visual anomaly cues, but do not explicitly analyze their relationship.

02 | Fusion ≠ Cross-Alignment - Multimodal fusion combines representations, but does not necessarily measure whether anomaly dynamics occur in a consistent temporal relationship.

03 | Sophisticated Lip-Sync - Modern generation can reproduce obvious phoneme–viseme synchronization.

→ Subtle cross-modal inconsistencies need to be examined.

04 | Computational Cost - Generator-aware methods provide strong forensic evidence but are too expensive to apply to every clip.

MOTIVATION

Need for an efficient approach that combines lightweight physiological evidence with explicit cross-modal alignment.

Problem Statement

Existing approaches face a trade-off

LIGHTWEIGHT	HEAVYWEIGHT
Physics-based audio / video cues	Generator-aware analysis
Low cost	High discrimination
Limited AV reasoning	High computation

Our Problem

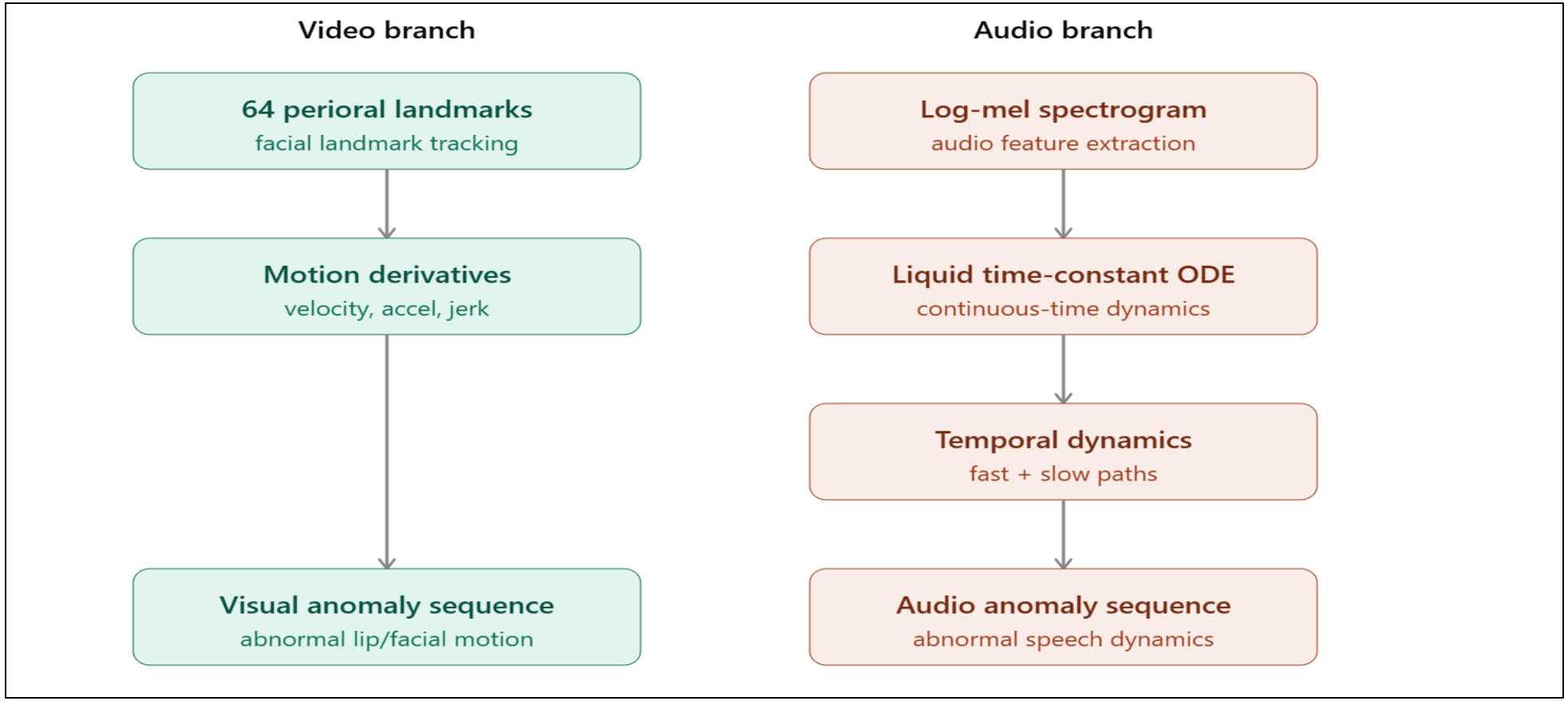
- **Develop an efficient multimodal deepfake detection pipeline** using audio and facial features.
- **Detect audio–facial inconsistencies** through explicit temporal cross-alignment.
- **Apply generator-aware analysis selectively** to ambiguous clips for improved efficiency.

Design Objective
Efficiency + Cross-modal reasoning + Strong evidence

Proposed Methodology

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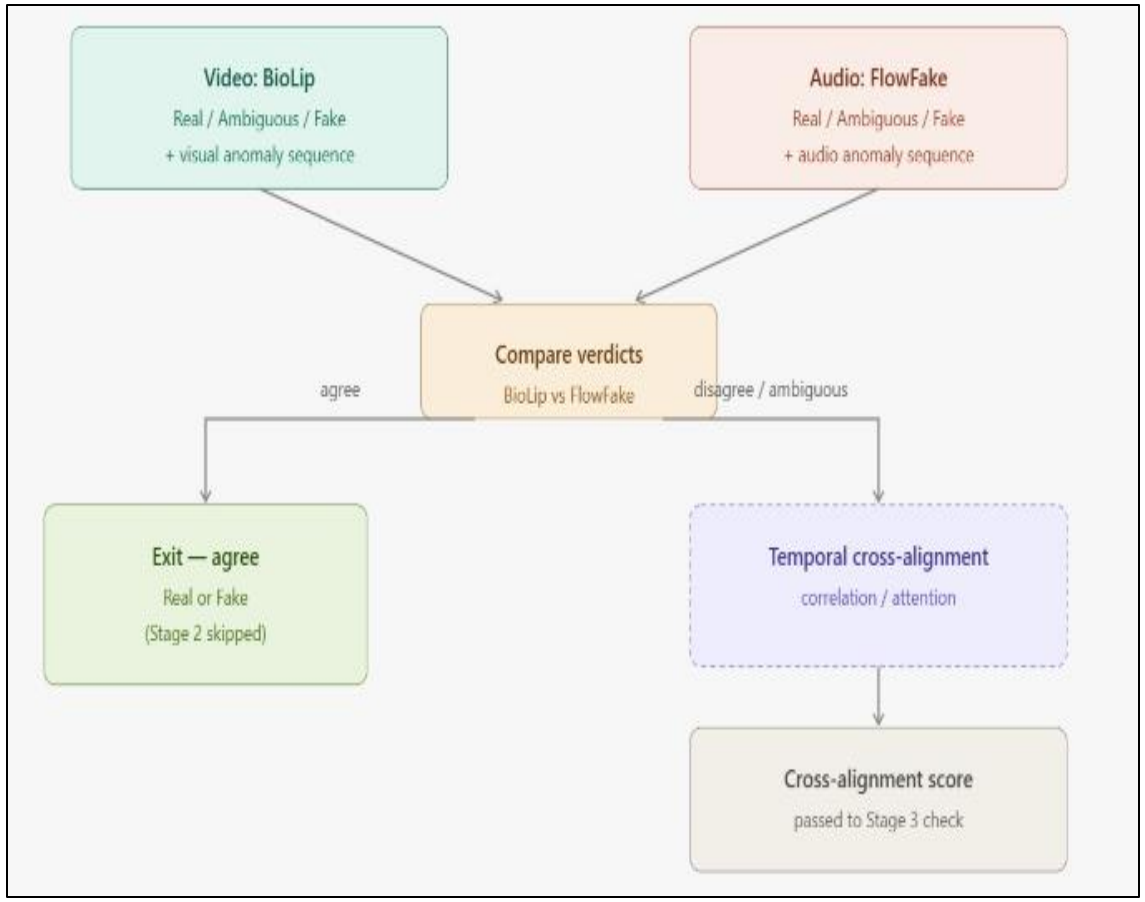
Stage 1 — Lightweight Anomaly Detection



Output: Two lightweight, modality-independent anomaly trajectories

Proposed Methodology Contd.

Stage 2 — Audio-Visual Cross-Alignment



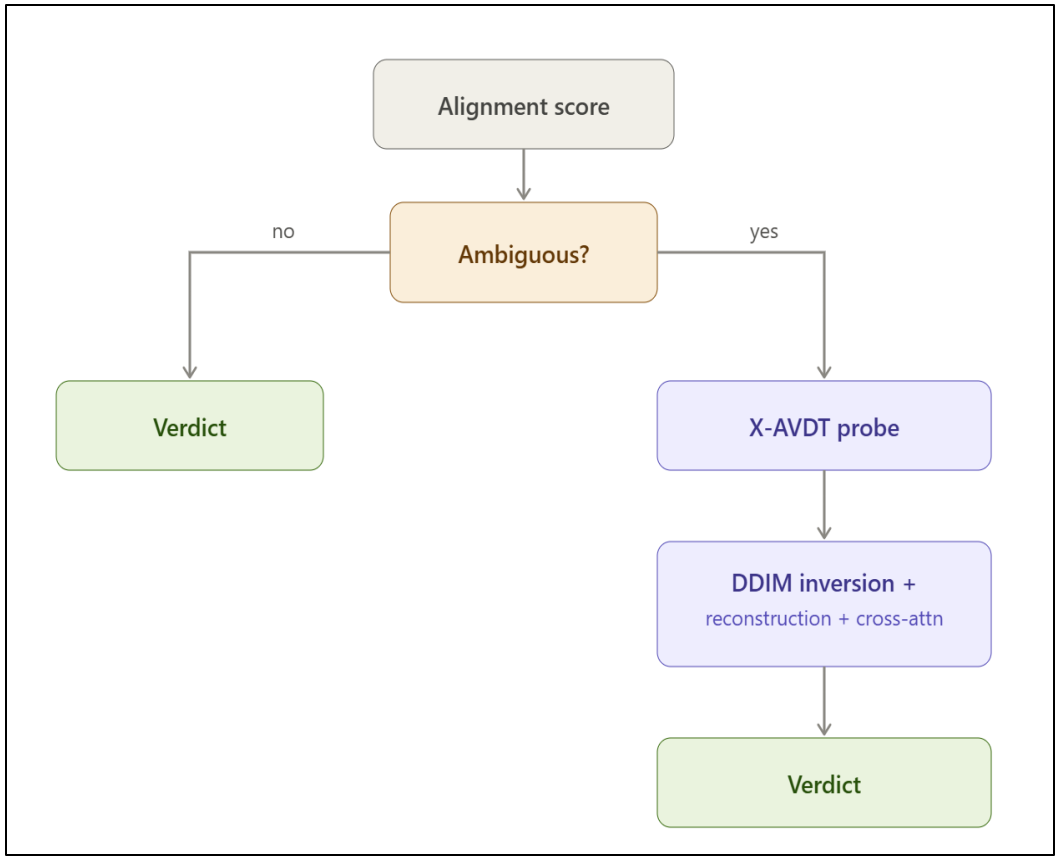
What is checked?

- **Temporal correspondence**
  Do anomalies occur at related times?
- **Coordination**
  Are audio and visual anomalies coupled?
- **Consistency**
  Are the two anomaly trajectories mutually compatible?

Align anomaly-score trajectories — not raw audio/video embeddings.

Proposed Methodology Contd.

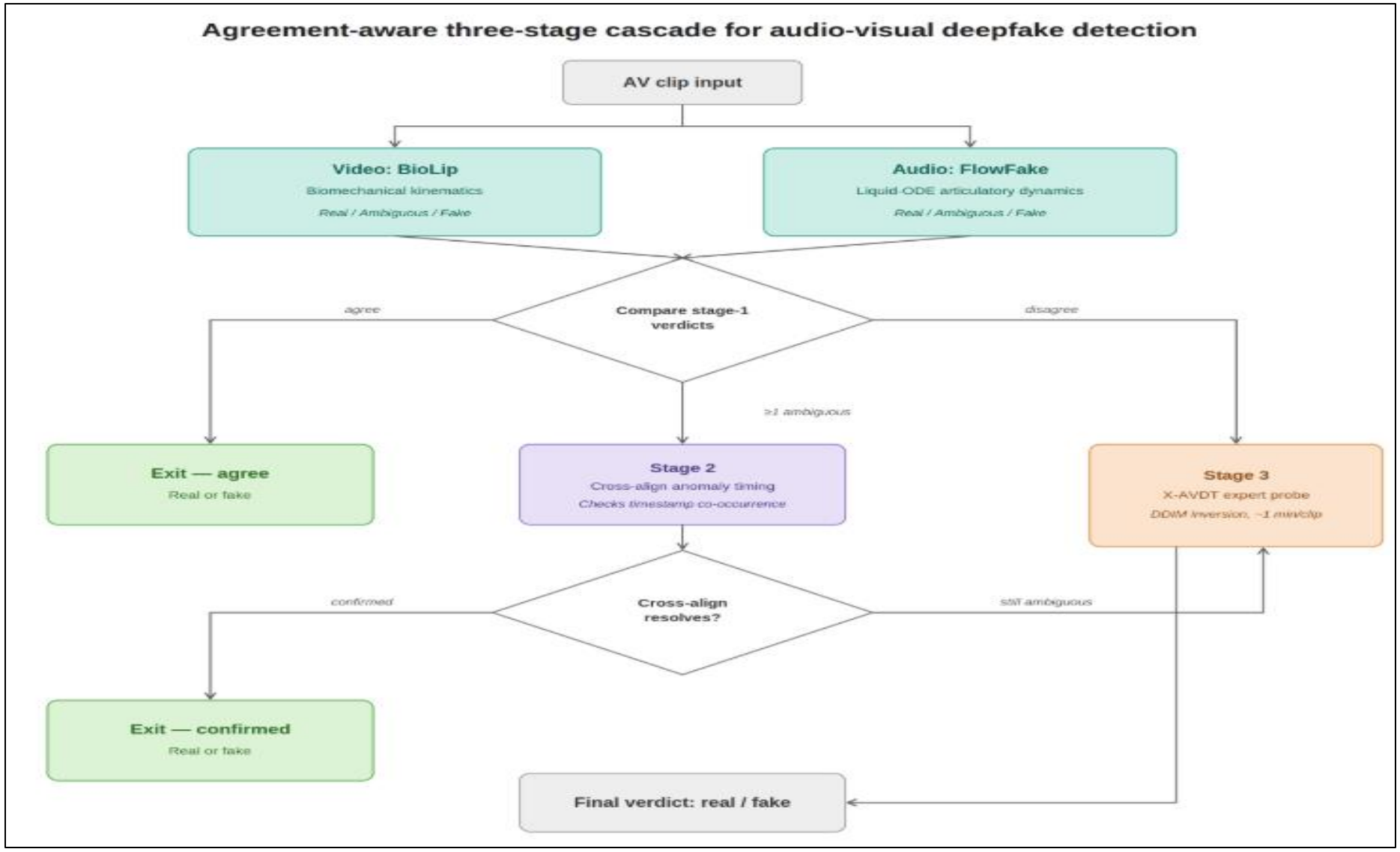
Stage 3 — Selective Generator-Aware Probing



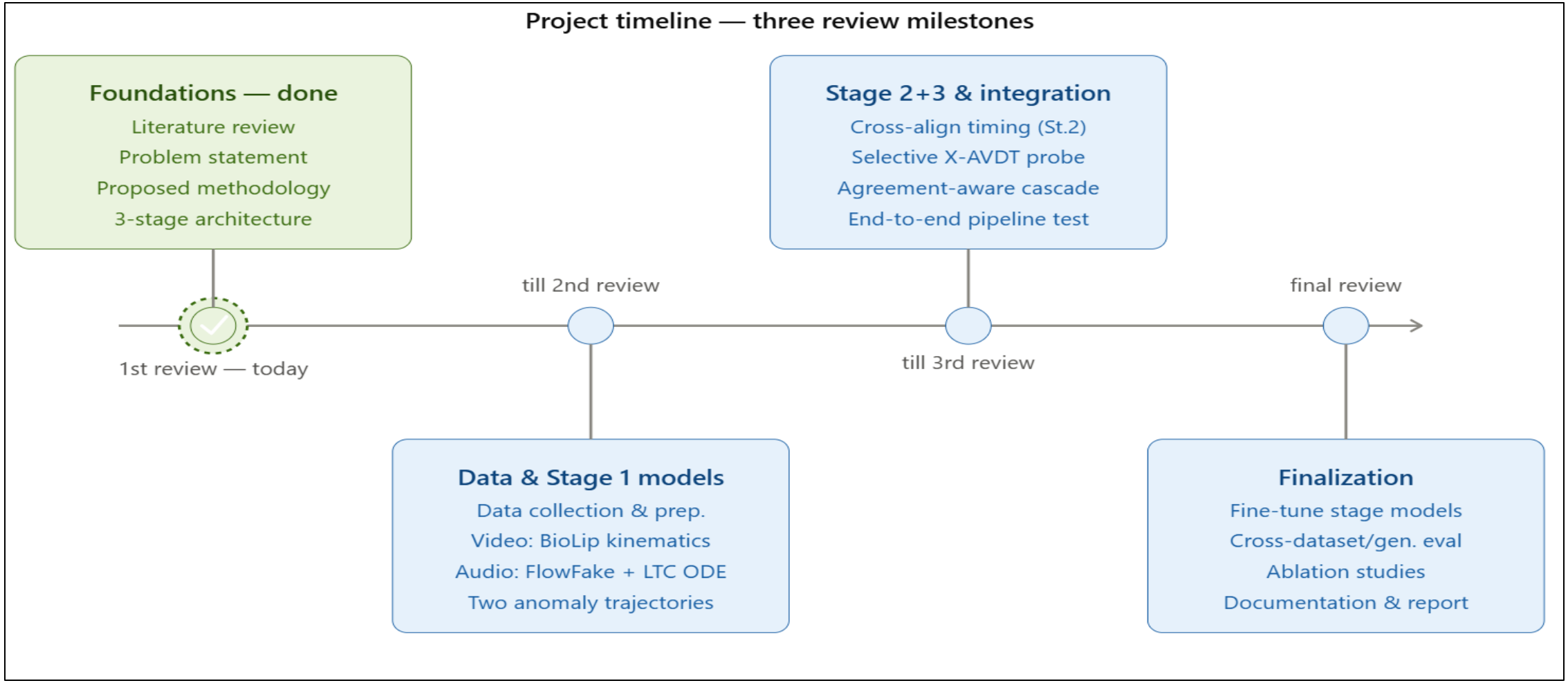
- Only **ambiguous clips** are **escalated**
- **Why selective?** - Stage 3 is computationally expensive.
- **Therefore**: Cheap evidence first → expensive analysis only when necessary

X-AVDT reports approximately 1 min/clip under its referenced diffusion-based setting.

Proposed Architecture



Future Workplan



Conclusion

- The proposed framework combines **audio and facial dynamics** to identify multimodal inconsistencies in deepfake videos.
- **Lightweight, modality-specific anomaly detectors** capture abnormal temporal patterns in speech and facial motion.
- An explicit **temporal cross-alignment module** evaluates whether these anomalies occur with the expected audio-visual coordination.
- **Generator-aware analysis** is selectively applied to ambiguous clips, reducing the need for expensive computation on every input.
- The overall approach aims to achieve a balance between **detection accuracy, interpretability, and computational efficiency**.

Key Principle:
Lightweight Anomaly Detection → Temporal Cross-Alignment → Selective Deep Analysis

References

[1] 'FlowFake - Liquid Networks for Audio Deepfake Detection ", 2026.

[2] "BioLip: Language-Generalizable Lip-Sync Deepfake Detection via Biomechanical Constraint Violation Modeling" , 2026.

[3] Oorloff et al., "AVFF: Audio-Visual Feature Fusion for Video Deepfake Detection," CVPR, 2024.

[4] "Cross-Modality and Within-Modality Regularization for Audio-Visual DeepFake Detection," ICASSP, 2024.

[5] "Contextual Cross-Modal Attention for Audio-Visual Deepfake Detection and Localization," 2024.

[6] "NPVForensics: Learning VA Correlations in Non-Critical Phoneme–Viseme Regions for Deepfake Detection."

[7] "AVSCNet: A Dual-Branch Network for Synchronization Detection and Content Consistency Learning in Audio-Video Forgery Detection," 2026.

[8] "X-AVDT: Audio-Visual Cross-Attention for Robust Deepfake Detection," CVPR, 2026.